

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.6 / 5.0
Confidence: 82%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	3.5	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	4.5	Excellent	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	2.5	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 95%

Strengths:

- [TITLE] is specific and informative, clearly reflecting the contribution of the study.
- [ABSTRACT] follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] include a mix of specific and general terms, with domain-specific indexing terms.

Minor Comments:

- [TITLE] could be slightly shortened for clarity without losing specificity.

Recommendations:

- [TITLE] Consider shortening the title slightly for improved readability while maintaining specificity.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear section mapping
- Sections are logically ordered with introduction, methodology, experiments, discussion, and conclusions

Weaknesses:

- Missing literature review section
- Section flow is disrupted by repeated content in 'discussion' and 'section_vi'
- Research problem and hypothesis are not clearly stated
- Objectives are not explicitly stated or matched with conclusions

Major Comments:

- The manuscript lacks a literature review section, which is essential for contextualizing the research problem and existing work.
- The repeated content in 'discussion' and 'section_vi' indicates a lack of coherence and logical flow.
- The research problem and hypothesis are not clearly defined in the introduction.
- The objectives are not explicitly stated and do not align with the conclusions.

Minor Comments:

- Section titles are not consistently formatted or descriptive.
- Some sections contain incomplete or truncated content, such as the introduction and conclusions.

Recommendations:

- Include a literature review section to provide context and justify the research problem.
- Revise the 'discussion' and 'section_vi' sections to eliminate redundancy and ensure logical flow.
- Clearly state the research problem, hypothesis, and objectives in the introduction.
- Ensure that the objectives are explicitly addressed in the conclusions.
- Complete and refine the content of all sections to ensure clarity and coherence.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 3.5 / 5.0 | Confidence: 80%

Strengths:

- Clear integration of physical constraints into the neural network framework
- Multi-objective optimization formulation
- Validation on a five-story building prototype

Weaknesses:

- No explicit description of the exact neural network architecture or training parameters
- No mention of the specific optimization algorithms used (e.g., genetic algorithm, particle swarm optimization) in detail
- No mention of the exact experimental setup or data preprocessing steps

Major Comments:

- The manuscript lacks detailed description of the neural network architecture and training parameters, which is critical for reproducibility.

Minor Comments:

- The experimental setup and data preprocessing steps should be described in more detail for reproducibility.

Recommendations:

- Include a detailed description of the neural network architecture, training parameters, and optimization algorithms used.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

Weaknesses:

- Lack of effect size reporting
- No verification of test assumptions (e.g., independence, homoscedasticity)
- No mention of how the 30 runs were justified or whether they are sufficient for the problem

Major Comments:

- The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and post-hoc Wilcoxon tests (e.g., independence, homoscedasticity) to ensure the validity of the results.
- Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the findings.

Minor Comments:

- The manuscript should clarify how the 30 independent runs were justified and whether they are sufficient for the problem's stochastic nature.
- Consider adding a section on the practical significance of the results, especially in the context of engineering applications.

Recommendations:

- Verify the assumptions of the Kruskal-Wallis and post-hoc tests (e.g., independence, homoscedasticity) using appropriate diagnostic tools (e.g., Shapiro-Wilk for normality, Levene's test for homoscedasticity).
- Report effect sizes (e.g., Cohen's d, eta-squared) for all significant findings to distinguish statistical from practical significance.
- Clarify the justification for using 30 independent runs and whether they are sufficient for the stochastic nature of the problem.

- Include a discussion on the practical significance of the results, especially in the context of engineering applications.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript includes a sufficient number of figures, tables, and equations to support the claims made in the text.
- Equations are sequentially numbered and appear to be properly introduced with context.
- Tables and figures are referenced in the text before they appear, which enhances clarity and coherence.

Weaknesses:

- The extracted text does not expose table bodies, so it is not possible to verify the completeness of table headers or data consistency with the text.

Major Comments:

- The manuscript should ensure that all tables and figures are fully described and their data is consistent with the claims made in the text.

Minor Comments:

- Consider adding more detailed captions for figures to ensure they are self-explanatory.

Recommendations:

- Include the full content of tables and figures in the manuscript for thorough evaluation.

ResultsReviewer

Results validity and empirical support

Score: 4.5 / 5.0 | Confidence: 90%

Strengths:

- Clear presentation of MSE, RMSE, CV, and hypervolume metrics
- Inclusion of Pareto analysis and baseline comparisons with GA and PSO
- Reporting of 30 independent runs for statistical robustness

Weaknesses:

- Lack of explicit mention of negative results or limitations of the EANN framework
- Incomplete description of the comparison with baseline methods (e.g., missing statistical significance tests)

Major Comments:

- The manuscript should explicitly report negative results or limitations of the EANN framework to ensure transparency.

Minor Comments:

- The description of the comparative analysis with GA and PSO is incomplete; more details on statistical significance tests are needed.

Recommendations:

- Include a section discussing limitations of the EANN framework, such as computational overhead or performance under certain conditions.
- Provide statistical significance tests (e.g., p-values, confidence intervals) to validate the performance differences between methods.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: 3.5 / 5.0 | Confidence: 80%

Strengths:

- The discussion provides a clear comparison of the EANN approach with GA and PSO in terms of performance metrics and statistical significance.
- The conclusions are directly tied to the experimental results and the hypothesis of the study.

Weaknesses:

- The discussion lacks a detailed comparison with prior work in the literature, which limits the contextualization of the findings.
- The limitations of the study are mentioned but not elaborated sufficiently.
- Future work directions are not clearly specified or justified.

Major Comments:

- The manuscript should include a more comprehensive comparison with existing literature to contextualize the findings and demonstrate the novelty of the EANN framework.

Minor Comments:

- The limitations section could be expanded to provide more depth on the constraints of the study.

Recommendations:

- Include a detailed comparison with prior work in the discussion to strengthen the contextualization of the results.

WritingReviewer

Scientific writing quality, clarity, and style

Score: 3.5 / 5.0 | Confidence: 80%

Strengths:

- Clear technical focus on physics-informed neural networks
- Well-structured sections with logical flow

Weaknesses:

- Overuse of passive voice
- Vague quantifiers without numerical support
- Unexplained acronyms

Major Comments:

- Passive voice overuse reduces clarity and engagement
- Acronyms like EANN should be defined on first use
- Vague terms like 'significant' weaken the precision of claims

Minor Comments:

- Some sentences are overly long and could be split for readability
- Technical terms should be consistently used throughout the manuscript

Recommendations:

- Define all acronyms on first use
- Revise passive voice constructions to active voice where possible
- Replace vague quantifiers with specific numerical values
- Split long sentences for improved readability

ReferencesReviewer

References quality, recency, and relevance

Score: 2.5 / 5.0 | Confidence: 70%

Strengths:

- Citations are consistent in style (IEEE format)

- Recent references (2021-2022) are included

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in the literature review
- Incomplete reference entries

Major Comments:

- The references section lacks seminal foundational works that are critical to the field of structural modeling and identification.
- Key areas such as structural health monitoring and seismic design regulations are not adequately represented in the references.
- The reference list is incomplete, with some entries cut off, which limits the confidence in the completeness of the literature review.

Minor Comments:

- The citation style is consistent with IEEE format.
- Recent references (2021-2022) are included, which is a positive aspect.

Recommendations:

- Include seminal foundational works in structural modeling and identification.
- Add references to key areas such as structural health monitoring and seismic design regulations.
- Complete the reference entries to ensure full visibility and accuracy.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Limitations are explicitly stated in the discussion section, indicating an awareness of the study's constraints.

Weaknesses:

- Limitations are not specific or detailed, which reduces their credibility and utility for readers.
- The manuscript does not address reproducibility, such as data and code availability, which is critical for scientific validation.

Major Comments:

- The manuscript lacks essential ethical reporting elements, including ethical approval, conflict of interest, funding disclosure, and data availability. These are required for compliance with COPE and IEEE standards.

Minor Comments:

- The limitations section is present but lacks specificity and depth, which weakens the credibility of the study's self-assessment.

Recommendations:

- Include a statement on ethical approval for any human or animal studies.
- Provide a data availability statement to ensure transparency and reproducibility.
- Declare any potential conflicts of interest.
- Disclose all funding sources that supported the research.
- Enhance the limitations section with specific, detailed information about the study's constraints and their impact on the conclusions.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript lacks a literature review section, which is essential for contextualizing the research problem and existing work.
- [2] The repeated content in 'discussion' and 'section_vi' indicates a lack of coherence and logical flow.
- [3] The research problem and hypothesis are not clearly defined in the introduction.
- [4] The objectives are not explicitly stated and do not align with the conclusions.
- [5] The manuscript lacks detailed description of the neural network architecture and training parameters, which is critical for reproducibility.
- [6] The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and post-hoc Wilcoxon tests (e.g., independence, homoscedasticity) to ensure the validity of the results.
- [7] Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the findings.
- [8] The manuscript should ensure that all tables and figures are fully described and their data is consistent with the claims made in the text.
- [9] The manuscript should explicitly report negative results or limitations of the EANN framework to ensure transparency.
- [10] The manuscript should include a more comprehensive comparison with existing literature to contextualize the findings and demonstrate the novelty of the EANN framework.
- [11] Passive voice overuse reduces clarity and engagement
- [12] Acronyms like EANN should be defined on first use

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] could be slightly shortened for clarity without losing specificity.
- [2] Section titles are not consistently formatted or descriptive.
- [3] Some sections contain incomplete or truncated content, such as the introduction and conclusions.
- [4] The experimental setup and data preprocessing steps should be described in more detail for reproducibility.
- [5] The manuscript should clarify how the 30 independent runs were justified and whether they are sufficient for the problem's stochastic nature.
- [6] Consider adding a section on the practical significance of the results, especially in the context of engineering applications.
- [7] Consider adding more detailed captions for figures to ensure they are self-explanatory.
- [8] The description of the comparative analysis with GA and PSO is incomplete; more details on statistical significance tests are needed.
- [9] The limitations section could be expanded to provide more depth on the constraints of the study.
- [10] Some sentences are overly long and could be split for readability
- [11] Technical terms should be consistently used throughout the manuscript
- [12] The citation style is consistent with IEEE format.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title slightly for improved readability while maintaining specificity.
2. Include a literature review section to provide context and justify the research problem.
3. Revise the 'discussion' and 'section_vi' sections to eliminate redundancy and ensure logical flow.

4. Clearly state the research problem, hypothesis, and objectives in the introduction.
5. Ensure that the objectives are explicitly addressed in the conclusions.
6. Complete and refine the content of all sections to ensure clarity and coherence.
7. Include a detailed description of the neural network architecture, training parameters, and optimization algorithms used.
8. Verify the assumptions of the Kruskal-Wallis and post-hoc tests (e.g., independence, homoscedasticity) using appropriate diagnostic tools (e.g., Shapiro-Wilk for normality, Levene's test for homoscedasticity).
9. Report effect sizes (e.g., Cohen's d, eta-squared) for all significant findings to distinguish statistical from practical significance.
10. Clarify the justification for using 30 independent runs and whether they are sufficient for the stochastic nature of the problem.
11. Include a discussion on the practical significance of the results, especially in the context of engineering applications.
12. Include the full content of tables and figures in the manuscript for thorough evaluation.